

Traditionally living populations in East Africa have a mean serum 25-hydroxyvitamin D concentration of 115 nmol/l.

Cutaneous synthesis of vitamin D by exposure to UVB is the principal source of vitamin D in the human body. Our current clothing habits and reduced time spent outdoors put us at risk of many insufficiency-related diseases that are associated with calcaemic and non-calcaemic functions of vitamin D. Populations with traditional lifestyles having lifelong, year-round exposure to tropical sunlight might provide us with information on optimal vitamin D status from an evolutionary perspective. We measured the sum of serum 25-hydroxyvitamin D α and D β (25(OH)D) concentrations of thirty-five pastoral Maasai (34 (SD 10) years, 43 % male) and twenty-five Hadzabe hunter-gatherers (35 (SD 12) years, 84 % male) living in Tanzania. They have skin type VI, have a moderate degree of clothing, spend the major part of the day outdoors, but avoid direct exposure to sunlight when possible. Their 25(OH)D concentrations were measured by liquid chromatography-MS/MS. The mean serum 25(OH)D concentrations of Maasai and Hadzabe were 119 (range 58-167) and 109 (range 71-171) nmol/l, respectively. These concentrations were not related to age, sex or BMI. People with traditional lifestyles, living in the cradle of mankind, have a mean circulating 25(OH)D concentration of 115 nmol/l. Whether this concentration is optimal under the conditions of the current Western lifestyle is uncertain, and should as a possible target be investigated with concomitant appreciation of other important factors in Ca homeostasis that we have changed since the agricultural revolution.